Fiononana RASAMIMANANA

Machine Learning Engineer & Geomatics Expert, Environment, Climate & Natural Risks, Remote Sensing, Deep Learning, GIS

CONTACT DETAILS

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EXPERIENCE

Date	Intern in Remote Sensing, Geomatics and Artificial Intelligence at the French National Institute for Industrial Environment and Risks (INERIS)	Verneuil-en-Halatte, France
Feb 2026 – Jul 2026	▪ Designed and implemented an end-to-end Deep Learning pipeline for the automatic detection of individual vegetable gardens, allotment gardens and ornamental gardens (trained on Île-de-France, tested on the Oise). ▪ Processed and prepared BD ORTHO® orthophotographs (IGN, 20 cm resolution) and computed spectral indices (NDVI, MSAVI2) used as additional model inputs. ▪ Built the training datasets : ground-truth masks, tiling, normalization and data augmentation (3,330 tiles of 512 × 512 pixels). ▪ Compared Machine Learning (Random Forest) and Deep Learning (U-Net with ResNet-34 encoder) models : U-Net reached 83 % overall accuracy versus 58 % for Random Forest ; on an Oise test area, IoU of 82 % (allotment gardens) and 56 % (individual vegetable gardens). ▪ Structured and managed a geospatial road network database in PostgreSQL / PostGIS. ▪ Developed Python scripts automating the collection and harmonization of open road traffic data at the departmental level (94 of 101 departments inventoried, 291,460 road segments).	

Aug 2024 – Jun 2025	**Intern in Geomatics, Remote Sensing and Environment at the Institute and Observatory of Geophysics of Antananarivo (IOGA), as part of the project funded by the Critical Ecosystem Partnership Fund (CEPF)**	Antananarivo, Madagascar

- Produced land cover maps of Ankarafantsika National Park and its surroundings from Sentinel-2 imagery for 2018 and 2023 (17 land cover classes).
- Classified land cover (tree-cover approach combined with Random Forest) and analysed land cover changes using remote sensing and field data.
- Assessed classification accuracy using confusion matrices, with overall accuracies of 91.4 % in 2018 and 93.0 % in 2023.
- Post-processed land cover maps in QGIS.
- Implemented an approach to natural capital ecosystem accounting covering land cover, functional ecosystem services and water resources in SAGA GIS (System for Automated Geoscientific Analyses).
- Analysed ecosystem dynamics and produced spatial indicators related to landscape fragmentation, resource accessibility and ecosystem health, showing a ∼1 % yearly decline in ecosystem infrastructure potential and a 4.4 % drop in net ecosystem water balance.
- Organized the August 2024 field mission and collected reference GPS points for the classification.

TECHNICAL SKILLS

Domain	Skills
Remote Sensing **&** Geomatics	▪ Processing of very high spatial resolution aerial imagery (IGN BD ORTHO, 20 cm resolution); ▪ Satellite image processing (Sentinel-1 and 2); ▪ Supervised and unsupervised classification; ▪ Machine Learning (Random Forest) and Deep Learning (U-Net) applications; ▪ Terrain model data analysis (IGN LiDAR digital height model, 50 cm resolution; NASA Shuttle Radar Topography Mission (SRTM) near-global elevation data, 30 m resolution); ▪ Thematic mapping (land cover, ecosystems); Spatial analysis; Web mapping (Python, Folium); Spatial database management (PostgreSQL / PostGIS); ▪ **Tools mastered** : QGIS, SNAP, Google Earth Engine (GEE), SAGA, OTB toolbox; PostgreSQL.
Geophysics	Knowledge of geophysical prospecting (electrical, magnetic and seismic methods).
Programming & Analysis	▪ Numerical modelling and analysis; Scripting to automate geospatial processing; Strong foundation in statistical learning; ▪ **Tools mastered** : Python (geospatial libraries: rasterio, NumPy, scikit-learn, Albumentations, Folium; deep learning on Google Colab GPU), SQL, Git, JavaScript (Google Earth Engine), MATLAB, SageMath; R (beginner).
Ecology & Fieldwork	▪ Knowledge of plant ecology and tropical ecosystems; field data collection with KoboCollect.

EDUCATION AND DEGREES

2026 – 2027	MSc in Applied Mathematics (in progress), Stellenbosch University, South Africa
2025 – 2026	International credit mobility in Natural Resources and Risks of Tropical Environments (RNET), specialization in Geomatics and Remote Sensing
	University of La Réunion, Faculty of Science and Technology
2024 – 2025	Master 2 (equivalent to MSc) in Sciences and Technologies, specialization in Geomatics and Remote Sensing
	University of Antananarivo, Faculty of Sciences, Earth Physics
	Master thesis (defended Aug 2026): Automatic detection of vegetable gardens by remote sensing and AI
2021 – 2022	Bachelor's degree in Physics and Applications
	University of Antananarivo, Faculty of Sciences
2017 – 2018	Baccalauréat (Series D), with Honours, Nanisana High School, Antananarivo

LANGUAGES & INTERESTS

Malagasy	Native language
French	Reading, speaking, writing (intermediate level)
English	Reading, speaking, writing (upper-intermediate, level III)
Russian	Reading, writing
Interests	Member of the Antananarivo Scientific Student Association (FMSA)
	Member of the American English course at IOGA; member of the Alliance Française of Antananarivo